# The Quotient Transversality Theorem
# A Proof-First Reduced Evans Identity for the Remaining DTFE Landing Obstructions

James Alan Darke

2026

**Abstract**

We isolate and prove a general reduced-Evans transversality identity for one-dimensional regular–admissible landing problems arising after symmetry reduction of self-adjoint Sturm–Liouville systems on the half-line. The result is designed to address the exact structural bottleneck now shared by the frozen DTFE hedgehog nondegeneracy problem and the grand-spin $K = 1$ reduced obstruction problem.

The theorem is proof-first and exact in scope. It does not prove nonvanishing of the remaining DTFE obstructions. What it proves is sharper at the structural level. If a reduced regular branch lands in the corresponding admissible decaying branch at a parameter value, then that landing is governed by an explicit crossing-form identity. In the spectral case $\mathcal{L}_\lambda = \mathcal{L}_0 - \lambda W$, the derivative of the reduced Evans pairing at a landing point is the negative weighted square norm of the landing mode. Hence any such landing is simple whenever the reduced weight is positive on nonzero quotient modes.

This theorem therefore removes one specific remaining ambiguity: the shared DTFE obstructions, if they vanish at all, do not vanish by flat tangency. They cross simply. We then record the two intended applications. First, the theorem applies to the hedgehog Jacobi landing problem, where strict positivity had already been reduced to exclusion of the decaying threshold mode. Secondly, it applies to the reduced $K = 1$ obstruction after symmetry removal, where the full nonradial question had already been reduced to one scalar Evans-type pairing. The present note proves simplicity of any such landing. It does not yet prove that the corresponding obstruction is nonzero.

## 1 Introduction

The current DTFE proof chain has already performed the main reductions. In the frozen hedgehog sector, strict positivity of the Jacobi operator was reduced to one nondegeneracy question: whether the distinguished regular Jacobi branch lands in the decaying threshold class at infinity. In the first genuinely coupled nonradial sector, grand spin $K = 1$, the full reduced problem was forced down to a single scalar obstruction after explicit symmetry removal. In both cases, the unresolved difficulty is no longer discovery of the operator, kernel, or correct reduction. It is the final landing obstruction.

That shared bottleneck suggests that the right next theorem is not a new brute-force numerical transport, but a structural statement about the landing itself.

The purpose of the present note is therefore narrow and exact. We prove a reduced-Evans transversality theorem for self-adjoint half-line problems in which the regular quotient line and the admissible decaying quotient line are both one-dimensional. The theorem shows that if such a landing occurs, then the derivative of the reduced Evans pairing is given by an explicit

crossing-form integral. In the spectral case, that integral is the weighted square norm of the landing mode, with definite sign once the reduced weight is positive.

This has one immediate consequence. Any remaining reduced landing obstruction in this framework, if it vanishes at all, vanishes simply rather than by flat tangency.

That is the exact theorem-level gain of the note. It does *not* by itself prove nonvanishing of the frozen hedgehog obstruction or of the reduced $K = 1$ obstruction. It does prove that the residual obstruction is not hiding in a higher-order grazing event. The shared bottleneck, if real, is a simple crossing problem.

# 2 Reduced self-adjoint landing problems

## 2.1 Operator setting

Let $\mu$ be a real parameter near $\mu_0$. Consider a smooth family of self-adjoint matrix Sturm–Liouville operators on $(0, \infty)$,

$$\mathcal{L}_\mu u := -\big(P(r,\mu)u'(r)\big)' + Q(r,\mu)u(r), \qquad u : (0,\infty) \to \mathbb{R}^m, \tag{1}$$

with the following standing hypotheses:

i. $P(r,\mu) = P(r,\mu)^T$ is $C^1$ in $r$, smooth in $\mu$, and positive definite for all $r > 0$ in the parameter range under consideration.

ii. $Q(r,\mu) = Q(r,\mu)^T$ is continuous in $r$ and smooth in $\mu$.

iii. The origin boundary condition defines a distinguished one-dimensional regular line $\mathcal{R}(\mu)$.

iv. The infinity boundary condition defines a distinguished one-dimensional admissible decaying line $\mathcal{D}(\mu)$.

v. Both distinguished lines vary smoothly with $\mu$.

We choose smooth nonzero representatives

$$u_{\mathrm{reg}}(\cdot,\mu) \in \mathcal{R}(\mu), \qquad u_{\mathrm{dec}}(\cdot,\mu) \in \mathcal{D}(\mu),$$

satisfying

$$\mathcal{L}_\mu u_{\mathrm{reg}} = 0, \qquad \mathcal{L}_\mu u_{\mathrm{dec}} = 0.$$

## 2.2 Lagrange boundary form

For each fixed $\mu$, define the Lagrange boundary form

$$\Omega_\mu(r; u, v) := u(r)^T P(r,\mu) v'(r) - u'(r)^T P(r,\mu) v(r). \tag{2}$$

**Lemma 2.1** (Lagrange identity)**.** *For sufficiently smooth $u, v$,*

$$v^T \mathcal{L}_\mu u - u^T \mathcal{L}_\mu v = -\frac{\mathrm{d}}{\mathrm{d}r}\,\Omega_\mu(r; u, v). \tag{3}$$

*In particular, if $\mathcal{L}_\mu u = \mathcal{L}_\mu v = 0$, then $\Omega_\mu(r; u, v)$ is constant in $r$.*

*Proof.* A direct expansion gives

$$v^T[-(Pu')' + Qu] - u^T[-(Pv')' + Qv] = -u^T(Pv')' + v^T(Pu')'.$$

Since $Q = Q^T$, the $Q$-terms cancel. Expanding the remaining derivative gives exactly

$$-\frac{\mathrm{d}}{\mathrm{d}r}(u^T Pv' - (u')^T Pv).$$

That is the stated identity. □

### 2.3 Reduced Evans pairing

**Definition 2.2** (Reduced Evans pairing)**.** The reduced Evans pairing is

$$\mathcal{E}_q(\mu) := \Omega_\mu(r; u_{\mathrm{reg}}(\cdot, \mu), u_{\mathrm{dec}}(\cdot, \mu)), \tag{4}$$

which is independent of $r$ because both branches solve the same homogeneous equation.

**Proposition 2.3** (Landing criterion)**.** *At a parameter value $\mu$, one has*

$$\mathcal{E}_q(\mu) = 0$$

*if and only if the regular line and the admissible decaying line coincide:*

$$\mathcal{R}(\mu) = \mathcal{D}(\mu).$$

*Equivalently, there exists a nontrivial solution of $\mathcal{L}_\mu u = 0$ that is both regular at the origin and admissibly decaying at infinity.*

*Proof.* Since both lines are one-dimensional, $\mathcal{E}_q(\mu) = 0$ means the Wronskian pairing of the two chosen generators vanishes. For a second-order self-adjoint half-line problem, vanishing of the boundary form of two homogeneous solutions is equivalent to linear dependence. Hence the two distinguished lines coincide if and only if $\mathcal{E}_q(\mu) = 0$. □

*Remark* 2.4. The quantity $\mathcal{E}_q$ depends on the chosen nonzero normalisations of $u_{\mathrm{reg}}$ and $u_{\mathrm{dec}}$, but only up to multiplication by a nonvanishing smooth factor. Therefore the vanishing set of $\mathcal{E}_q$ and the simplicity or nonsimplicity of its zeros are invariant under such renormalisations.

## 3 The Quotient Transversality Theorem

We now state and prove the central identity.

**Theorem 3.1** (Parametric Quotient Transversality Theorem)**.** *Assume the setting of the previous section, and suppose that*

$$\mathcal{E}_q(\mu_0) = 0.$$

*Choose the representatives $u_{\mathrm{reg}}$ and $u_{\mathrm{dec}}$ so that at $\mu = \mu_0$,*

$$u_{\mathrm{reg}}(\cdot, \mu_0) = u_{\mathrm{dec}}(\cdot, \mu_0) =: u_0,$$

*with $u_0 \not\equiv 0$.*

*Assume further that the chosen regular normalisation is fixed at the origin and the chosen admissible normalisation is fixed at infinity, so that the corresponding parameter derivatives remain in the same regular and admissible boundary classes.*

*Then*

$$\mathcal{E}_q'(\mu_0) = \int_0^\infty u_0(r)^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu=\mu_0} u_0(r)\ \mathrm{d}r. \tag{5}$$

*In particular, if the quadratic crossing form*

$$\mathfrak{C}_{\mu_0}[u_0] := \int_0^\infty u_0(r)^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu=\mu_0} u_0(r)\ \mathrm{d}r \tag{6}$$

*is nonzero, then the landing at $\mu_0$ is simple:*

$$\mathcal{E}_q'(\mu_0) \neq 0.$$

*Proof.* Differentiate

$$\mathcal{L}_\mu u_{\mathrm{reg}}(\cdot,\mu) = 0, \qquad \mathcal{L}_\mu u_{\mathrm{dec}}(\cdot,\mu) = 0$$

with respect to $\mu$ at $\mu = \mu_0$. Writing dots for $\partial_\mu|_{\mu_0}$, we obtain

$$\mathcal{L}_{\mu_0}\dot{u}_{\mathrm{reg}} = -(\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0, \qquad \mathcal{L}_{\mu_0}\dot{u}_{\mathrm{dec}} = -(\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0. \tag{7}$$

Apply the Lagrange identity with $u = \dot{u}_{\mathrm{reg}}$ and $v = u_0$. Since $\mathcal{L}_{\mu_0} u_0 = 0$,

$$u_0^T \mathcal{L}_{\mu_0}\dot{u}_{\mathrm{reg}} = -\frac{\mathrm{d}}{\mathrm{d}r}\,\Omega_{\mu_0}(r;\dot{u}_{\mathrm{reg}},u_0).$$

Substituting the differentiated equation gives

$$-u_0^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0 = -\frac{\mathrm{d}}{\mathrm{d}r}\,\Omega_{\mu_0}(r;\dot{u}_{\mathrm{reg}},u_0),$$

hence

$$\frac{\mathrm{d}}{\mathrm{d}r}\,\Omega_{\mu_0}(r;\dot{u}_{\mathrm{reg}},u_0) = u_0^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0. \tag{8}$$

Integrating from 0 to $R$ gives

$$\Omega_{\mu_0}(R;\dot{u}_{\mathrm{reg}},u_0) - \Omega_{\mu_0}(0;\dot{u}_{\mathrm{reg}},u_0) = \int_0^R u_0^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0\ \mathrm{d}r. \tag{9}$$

Because the regular normalisation is fixed at the origin, the origin boundary term vanishes:

$$\Omega_{\mu_0}(0;\dot{u}_{\mathrm{reg}},u_0) = 0.$$

Therefore

$$\Omega_{\mu_0}(R;\dot{u}_{\mathrm{reg}},u_0) = \int_0^R u_0^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0\ \mathrm{d}r. \tag{10}$$

Now differentiate the reduced Evans pairing:

$$\mathcal{E}_q'(\mu_0) = \Omega_{\mu_0}(r;\dot{u}_{\mathrm{reg}},u_0) + \Omega_{\mu_0}(r;u_0,\dot{u}_{\mathrm{dec}}) + (\partial_\mu \Omega_\mu)\big|_{\mu_0}(r;u_0,u_0).$$

The last term vanishes because $\Omega_\mu(\cdot;\,\cdot,\cdot)$ is skew in the solution arguments for every $\mu$, hence

$$\Omega_\mu(r;u_0,u_0) \equiv 0 \quad \text{for all } \mu.$$

Also,

$$\Omega_{\mu_0}(r;u_0,\dot{u}_{\mathrm{dec}}) = -\Omega_{\mu_0}(r;\dot{u}_{\mathrm{dec}},u_0).$$

Thus

$$\mathcal{E}_q'(\mu_0) = \Omega_{\mu_0}(r;\dot{u}_{\mathrm{reg}},u_0) - \Omega_{\mu_0}(r;\dot{u}_{\mathrm{dec}},u_0). \tag{11}$$

This expression is independent of $r$. Evaluate it in the limit $r \to \infty$. Since the admissible normalisation is fixed at infinity, one has

$$\Omega_{\mu_0}(\infty; \dot{u}_{\mathrm{dec}}, u_0) = 0.$$

Hence

$$\mathcal{E}_q'(\mu_0) = \lim_{R\to\infty} \Omega_{\mu_0}(R; \dot{u}_{\mathrm{reg}}, u_0).$$

Using the integrated identity above and passing to the limit $R \to \infty$ gives

$$\mathcal{E}_q'(\mu_0) = \int_0^\infty u_0(r)^T (\partial_\mu \mathcal{L}_\mu)\big|_{\mu_0} u_0(r)\ \mathrm{d}r.$$

This is exactly the stated identity. □

**Proposition 3.2** (Normalisation invariance)**.** *Let*

$$\widetilde{u}_{\mathrm{reg}}(\cdot, \mu) = \alpha(\mu)\, u_{\mathrm{reg}}(\cdot, \mu), \qquad \widetilde{u}_{\mathrm{dec}}(\cdot, \mu) = \beta(\mu)\, u_{\mathrm{dec}}(\cdot, \mu),$$

*with $\alpha, \beta$ smooth and nonvanishing near $\mu_0$. Then*

$$\widetilde{\mathcal{E}}_q(\mu) = \alpha(\mu)\beta(\mu)\mathcal{E}_q(\mu).$$

*In particular, if $\mathcal{E}_q(\mu_0) = 0$, then*

$$\widetilde{\mathcal{E}}_q'(\mu_0) = \alpha(\mu_0)\beta(\mu_0)\mathcal{E}_q'(\mu_0),$$

*so the property $\mathcal{E}_q'(\mu_0) \neq 0$ is independent of the chosen nonzero normalisations.*

*Proof.* The boundary form is bilinear in the two solution arguments. The first identity follows immediately. At a zero of $\mathcal{E}_q$, differentiation yields the second identity because the terms involving $\alpha'$ and $\beta'$ are multiplied by $\mathcal{E}_q(\mu_0) = 0$. □

# 4 Spectral corollary

The most important special case is the spectral pencil

$$\mathcal{L}_\lambda = \mathcal{L}_0 - \lambda W(r), \tag{12}$$

where $W(r) = W(r)^T$ is a symmetric matrix weight.

**Corollary 4.1** (Spectral Quotient Transversality)**.** *Assume the hypotheses of the theorem for the spectral family*

$$\mathcal{L}_\lambda = \mathcal{L}_0 - \lambda W.$$

*Let $\lambda_0$ be a reduced landing point, and let $u_0$ be the corresponding nontrivial regular–admissible quotient mode. Then*

$$\mathcal{E}_q'(\lambda_0) = -\int_0^\infty u_0(r)^T W(r) u_0(r)\ \mathrm{d}r. \tag{13}$$

*In particular, if $W$ is positive on nonzero quotient modes, then*

$$\mathcal{E}_q'(\lambda_0) \neq 0.$$

*Hence every reduced spectral landing is simple.*

*Proof.* One has

$$\partial_\lambda \mathcal{L}_\lambda = -W.$$

Substituting into the theorem gives the identity immediately. □

*Remark* 4.2. The global sign of $\mathcal{E}_q'(\lambda_0)$ depends only on the chosen Evans orientation. The theorem-level content is not the sign convention itself but the nonvanishing. A landing, if it occurs, is a simple crossing rather than a flat tangency.

# 5 Application to the frozen hedgehog Jacobi obstruction

## 5.1 Jacobi operator and threshold landing problem

In the frozen DTFE hedgehog sector, the second variation is governed by the Jacobi operator

$$J_{f_*}\eta = -2\frac{\mathrm{d}}{\mathrm{d}r}\Big((r^2 + 2\sin^2 f_*)\eta'(r)\Big) + V_{\mathrm{eff}}(r)\eta(r), \tag{14}$$

and strict positivity had already been reduced to one remaining question: whether the distinguished regular Jacobi branch lands in the decaying zero-mode class at infinity.

Equivalently, in the shifted spectral family

$$J_{f_*}\eta = \lambda\eta, \tag{15}$$

the unresolved threshold problem is whether there exists a nontrivial solution that is both regular at the origin and decaying at infinity when $\lambda = 0$.

## 5.2 Reduced Evans interpretation

For the scalar hedgehog Jacobi problem, the reduced setting is one-dimensional from the start. The regular line at the origin is generated by the derivative of the local shooting family, and the admissible decaying line at infinity is the $r^{-2}$ threshold class.

The previously isolated obstruction may be read equivalently in either of the following two forms:

i. vanishing of the reduced Wronskian pairing between the regular branch and the decaying threshold branch;

ii. vanishing of the asymptotic growing coefficient $A(\infty)$ in the large-radius form

$$\eta(r) = A(\infty)\, r + B(\infty)\, r^{-2} + o(r^{-2}).$$

These two formulations differ only by a nonzero normalisation factor.

**Proposition 5.1** (Simplicity of any hedgehog threshold landing)**.** *Suppose the frozen hedgehog Jacobi problem admits a nontrivial threshold mode $\eta_0$ that is regular at the origin and decays at infinity. Then the corresponding reduced Evans zero at $\lambda = 0$ is simple. More precisely,*

$$\mathcal{E}_q'(0) = -\int_0^\infty \eta_0(r)^2 \,\mathrm{d}r < 0, \tag{16}$$

*up to the fixed global Evans orientation convention.*

*Proof.* This is the scalar case of the spectral corollary with

$$W(r) \equiv 1.$$

Since $\eta_0 \not\equiv 0$, the integral is strictly positive, and therefore $\mathcal{E}_q'(0) \neq 0$. □

*Remark* 5.2. This proposition does not prove that the hedgehog obstruction vanishes or does not vanish. It proves the narrower structural fact that if a threshold landing occurs, it occurs transversely rather than by flat tangency. In particular, the remaining frozen hedgehog obstruction is not a higher-order grazing event.

# 6 Application to the grand-spin $K = 1$ reduced obstruction

## 6.1 Reduced $K = 1$ structure

For the frozen DTFE parent action, the first genuinely coupled nonradial block is the grand-spin $K = 1$ sector. The exact reduced analysis already isolates:

i. the explicit three-channel radial system in the true orbital basis ($L = 0, 1, 2$);

ii. the positive channel metric
$$M = \mathrm{diag}\left(1, \frac{2}{3}, 2\right);$$

iii. the exact symmetry kernel generated by translation and admissible global orientation;

iv. the canonical regular non-symmetry branch $R_2$;

v. the canonical remaining admissible decaying branch $D_0$;

vi. the scalar obstruction
$$\Theta_{K=1} = W(R_2, D_0),$$
where $W$ is the conserved covariant Wronskian of the reduced self-adjoint system.

After symmetry removal, the remaining regular quotient line and the remaining admissible decaying quotient line are both one-dimensional. The unresolved question is whether these two quotient lines coincide.

## 6.2 Reduced spectral continuation

Let the reduced $K = 1$ system be continued to a local spectral family in $\lambda$, and let $\mathcal{E}_{K=1}(\lambda)$ denote the corresponding reduced Evans pairing between:

i. the distinguished continuation of the canonical regular quotient branch issued from the $r^2 C_2$ class at the origin;

ii. the distinguished continuation of the canonical admissible quotient branch issued from the $r^{-1} C_0$ class at infinity.

At $\lambda = 0$, this reduced Evans pairing agrees with the previously isolated obstruction $\Theta_{K=1}$, up to a fixed nonzero normalisation factor.

**Proposition 6.1** (Conditional simplicity of any $K = 1$ quotient landing)**.** *Assume that the local spectral continuation of the reduced $K = 1$ quotient problem is self-adjoint and carries a positive quotient spectral weight $W_{K=1}$ on nonzero quotient modes. Suppose that at $\lambda = 0$ there exists a nontrivial reduced $K = 1$ mode $u_0$ that is both regular at the origin and admissibly decaying at infinity after symmetry removal. Then the corresponding reduced Evans zero is simple. More precisely,*
$$\mathcal{E}'_{K=1}(0) = -\int_0^\infty u_0(r)^T W_{K=1}(r) u_0(r)\, \mathrm{d}r, \tag{17}$$
*and therefore*
$$\mathcal{E}'_{K=1}(0) \neq 0.$$
*Equivalently, any zero of $\Theta_{K=1}$ is a simple quotient crossing rather than a flat quotient tangency.*

*Proof.* This is an immediate application of the spectral corollary to the reduced one-dimensional quotient landing problem, once the positive quotient weight is available. □

*Remark* 6.2. The extra hypothesis here is exact and should not be hidden. The present note proves the reduced transversality mechanism. To convert that mechanism into a strict $K = 1$ spectral crossing statement, one needs the reduced quotient spectral weight written down explicitly and shown positive on nonzero quotient modes. The theorem is therefore structurally ready for the $K = 1$ problem, but not claimed here as full closure of the $K = 1$ nonvanishing question.

# 7 Boundary of proven scope

The exact force of the present note is the following.

1. It proves a reduced-Evans derivative identity for one-dimensional regular–admissible quotient landing problems in self-adjoint half-line systems.
2. In the spectral case, it proves that any such landing is simple once the reduced weight is positive on nonzero quotient modes.
3. It therefore proves that the remaining frozen DTFE landing obstructions, if they vanish at all, do not vanish by flat tangency.

What the note does *not* prove is equally important.

1. It does not prove that the frozen hedgehog threshold obstruction is nonzero.
2. It does not prove that the reduced $K = 1$ obstruction $\Theta_{K=1}$ is nonzero.
3. It does not supply the missing finite-radius anchor or the final analytic tail exclusion.
4. It does not replace the existing hedgehog or $K = 1$ papers. It supplements them by hardening one specific structural point in the remaining proof chain.

So the claim boundary is exact: the present theorem settles simplicity of any reduced landing, not nonvanishing of the obstruction itself.

# 8 Conclusion

The remaining frozen DTFE proof bottleneck is no longer discovery of the correct reduced objects. Those objects are already known. In the hedgehog sector, the problem had already been reduced to one threshold landing obstruction. In the grand-spin $K = 1$ sector, the problem had already been reduced to one scalar quotient obstruction after explicit symmetry removal.

The present note proves a structural theorem exactly matched to that stage. In any self-adjoint one-dimensional regular–admissible quotient landing problem, the derivative of the reduced Evans pairing at a landing point is given by an explicit crossing-form integral. In the spectral case, this is the negative weighted square norm of the landing mode. Therefore any such landing is simple whenever the reduced weight is positive.

This theorem does not yet complete the frozen DTFE nondegeneracy chain. It does, however, remove one precise ambiguity. The remaining obstructions, if they vanish at all, are simple crossings rather than flat tangencies. The proof chain is therefore harder and cleaner than before, even though the final nonvanishing step remains open.

## References


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